REC 5: Breaking Changes and Semantic Incompatibilities

Overview

REC 5 introduces significant semantic changes that break backward compatibility with previous REC versions. This document summarizes the major incompatibilities and migration considerations.

1. Complete ICT Equipment Hierarchy Removal

Impact: 21 ICT-related classes completely removed from REC domain

Removed Classes:

- `ICTEquipment` (root class) and `ICTHardware`
- All networking equipment: `DataNetworkEquipment` , `EthernetPort` , `EthernetSwitch` , `NetworkRouter` , `NetworkSecurityEquipment` , `WirelessAccessPoint`
- All controllers: `Controller` , `BACnetController` , `ModbusController` , `Gateway`
- Server infrastructure: `Server` , `ITRack`
- All sensor equipment: `SensorEquipment` , `DaylightSensorEquipment` , `IAQSensorEquipment` , `LeakDetectorEquipment` , `OccupancySensorEquipment` , `PeopleCountSensorEquipment` , `ThermostatEquipment` , `VibrationSensorEquipment`
- Audio/visual: `AudioVisualEquipment`
- Connectivity parameters: `GatewayConnectionParameter` , `IoTHubConnectionParameter`

Migration Path: Use Brick Schema for all ICT/technical equipment

2. Excluded Properties and Relationships

Custom Properties Removed:

- `customProperties` and `customTags` excluded from all core classes (`Agent` , `Asset` , `BuildingElement` , `Collection` , `Event` , `Information` , `Space`)

Substance Relationships Removed:

- `Architecture.isFedBy` excluded (use Brick substance classes instead)

Missing Enumeration Properties:

- `EthernetPort.poeType` and `WirelessAccessPoint.generation` removed with ICT equipment

3. Observation Events Restructuring

Removed Observation Subclasses (41 classes):

- All specific observation types (`TemperatureObservation` , `PressureObservation` , etc.)
- Only generic `ObservationEvent` and `PointEvent` remain

Impact: Applications using specific observation types must migrate to Brick Schema's `brick:Point` with `brick:lastKnownValue`

4. Open-Ended Relationship Constraints

Properties with Intentionally Open Ranges:

- `Agent.owns` - can own any entity (REC, Brick, or external ontologies)
- `Document.documentTopic` - can reference any entity type
- `PointOfInterest.objectOfInterest` - open for cross-ontology references
- `ServiceObject.relatedTo` - unrestricted relationship target

Impact: Less type safety but greater interoperability with external ontologies

5. Union Type Handling Changes

Simplified Union Ranges:

- `Portfolio.includes` : `rec:Architecture | rec:Collection` → uses first type for SHACL validation
- `Lease.leaseOf` : `rec:Collection | rec:Space` → uses first type for SHACL validation
- `Space.isLocationOf` : `rec:Asset | rec:BuildingElement` → uses first type for SHACL validation

Impact: SHACL validation may be less restrictive than intended union semantics

6. Domain Strategy Changes

Minimal Domain Approach:

- Most properties lack explicit OWL domains
- Only signature properties (like `hasPoint`) have domains
- SHACL shapes provide domain constraints instead of OWL

Impact: More flexible querying but less OWL reasoning capability

7. Translation Configuration Dependencies

New Requirements:

- Applications must now configure translation decisions
- Enumeration strategies must be explicitly defined
- Missing ranges require configuration or remain open
- Swedish translations optional but configured by default

Migration Recommendations

1. **ICT Equipment**: Migrate all technical equipment models to Brick Schema
2. **Custom Properties**: Implement custom properties through separate extension ontologies
3. **Observations**: Replace specific observation classes with Brick Points and `lastKnownValue`
4. **Open Ranges**: Review integration points for cross-ontology relationships
5. **Validation**: Update SHACL validation expectations for simplified union types
6. **Configuration**: Implement translation configuration management for controlled deployments

Compatibility Assessment

High Impact Changes:

- Complete ICT equipment removal
- Observation event restructuring
- Custom property exclusions

Medium Impact Changes:

- Open-ended relationship ranges
- Union type simplification
- Domain strategy changes

Low Impact Changes:

- Translation configuration requirements
- Swedish translation additions

This document reflects REC 5 as configured with the comprehensive ICT equipment migration to Brick Schema and systematic translation decision implementation.